Table 2 Data relevant to *p*H spectra in Fig. 3

Spectrum	Wind speed ($m\ s^{-1}$)	Rainfall rate* ($mm\ h^{-1}$)	Start† (min)	Duration (min)	*p*H variation‡	Bulk *p*H
1	1.1±0.2	4±0.5	40	40	0.06	5.50±0.02
2	0.5±0.2	7±0.5	30	30	0.07	5.54±0.02
3	1.6±0.3	5±0.5	20	30	0.12	5.38±0.02
4	1.4±0.2	5±0.5	24	35	0.13	5.44±0.02
5	0.7±0.3	7±0.5	0	22	0.29	5.30±0.02
6	1.9±0.3	8±0.5	0	24	0.43	5.30±0.02

* Rainfall rate derived from water volume collected by spectrometer.
† Time of start of sampling relative to start of rain shower.
‡ Standard deviation computed from the six *p*H spectral points in each spectrum.

all airmasses were comparable. These data suggest that the time constant for spectrum change is ~20 min.

These results are the first test of the drop-size dependence predicted by acid-rain scavenging models; laboratory wind-tunnel experiments are not guaranteed to simulate natural conditions and bulk-water measurements do not allow detailed microphysical inferences. For example, the maritime airmass examples all have comparable bulk-water *p*H values near the CO_2 equilibrium, but when size-fractionated results are plotted, *p*H varies by as much as 1.2 units.

The observed transient *p*H spectra agree qualitatively with present steady-state raindrop scavenging models assuming low NH_3 concentrations[1,3]. There is, however, a need for comparison between time-dependent models and data such as our instrument can obtain. Collection of rain over a wider range of conditions, with a more comprehensive error treatment and chemical analysis of other components, is also desirable to allow closer comparison between theory and measurement.

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Palaeolatitude drift history of displaced terranes in southern and Baja California

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The delineation of microplate-tectonic terranes[1] which make up western North America, the timing of their accretions to western North America and the identification of their original geological settings are all topics of debate[2,3]. The most recent addition to this family of terranes is the Santa Lucia/Orocopia allochthon[4,5] (SLOA) (Fig. 1), which is made up of Salinia[6,7] and several other smaller terranes[8] and is located in central and southern California. This composite terrane amalgamated in the Cretaceous and accreted to southern California by the early Palaeogene[4,5]. We describe here new palaeomagnetic results from localities outboard from the SLOA in southern California (USA) and Baja California (Mexico). These results are used to define a newly identified allochthonous terrane in southern and Baja California (Peninsular Ranges terrane[9], PRT) which is outboard from the SLOA, estimate the PRT's palaeolatitude drift history for the past 160 Myr, and infer its original geological setting. PRT and SLOA palaeolatitude drift histories are then combined to produce a new perspective on the tectonic development of southern and Baja California over the past 160 Myr.

Our palaeomagnetic results are based on samples collected from Palaeocene (Silverado Formation), Cretaceous (Ladd and Williams Formations) and Jurassic (Bedford Canyon Formation) sedimentary rocks of the Peninsular Ranges (southern California), and Cretaceous sedimentary rocks of the Santa Monica Mountains (southern California; Tuna Canyon Formation) and El Rosario (Baja California; Punta Baja and Rosario Formations). At each sampling locality, four oriented specimens were collected from each of several sites. The Cretaceous sampling localities and their relations to our proposed terrane boundaries are shown in Fig. 1 (see also Table 1).

All specimens were thermally demagnetized at 50 or 100 °C steps up to at least 500 °C. Zijderveldt diagrams (Fig. 2*a*) were used to estimate the characteristic remanences associated with the primary and any secondary magnetizations. Secondary magnetizations were common and easily removed by 200 °C. Characteristic remanences associated with the primary magnetization (that is, that remanence acquired at the time of initial sediment deposition) were recovered routinely between 200 and 500 °C. Rock magnetic analysis indicated that the primary magnetization was carried by detrital magnetite. If good characteristic remanences could not be discerned, the specimens were discarded from further analyses. Three criteria were then applied to the characteristic remanences to further establish that they were primary in origin. The criteria (see Table 1) were: (1) the presence of well-defined (antipodal polarity) reversals; (2) the passage of a fold test, and (3) *in situ* directions significantly different from the present axial-dipole field. At each of the sampling localities, either two or three of the criteria were satisfied. The results from the Palaeocene and Jurassic rocks of the Santa Ana Mountains should be considered more tentative than the Cretaceous results due to the smaller number of sampling sites at these two localities.

The resultant field vectors for each locality, before (*in situ*) and after bedding correction, are plotted in Fig. 2*b*, and listed in Table 1. The apparent similarity of most *in situ* average field vectors could be interpreted to mean that the rocks were remagnetized after folding while still at lower latitudes. However, the presence of well-defined intervals of reversed and normal polarity in all localities, the fact that the magnetostratigraphies correlate well with the expected magnetic polarity stratigraphy estimated from known biostratigraphical data[10], and the fact that three out of five localities clearly pass fold tests argue convincingly against such a possibility. More detailed analyses of the basic data are presented elsewhere (ref. 10 and L.K.M. and J. G. Fry, unpublished data).

The inclination results were used to estimate palaeolatitudes for each locality based on an axial-dipole mapping. Each locality's palaeolatitude was then normalized to its current latitude to produce the quantity $\Delta\theta$ (palaeolatitude minus present-day latitude) (Table 1).

Our normalized palaeolatitudes, $\Delta\theta$s, and their associated declinations are compared with other previously published results[5,7,8,11–13] of Cretaceous age in Fig. 1. This shows that all localities south of the Oak Ridge Fault (ORF) and its extension into Santa Barbara Basin have small negative $\Delta\theta s$ of about −6°.

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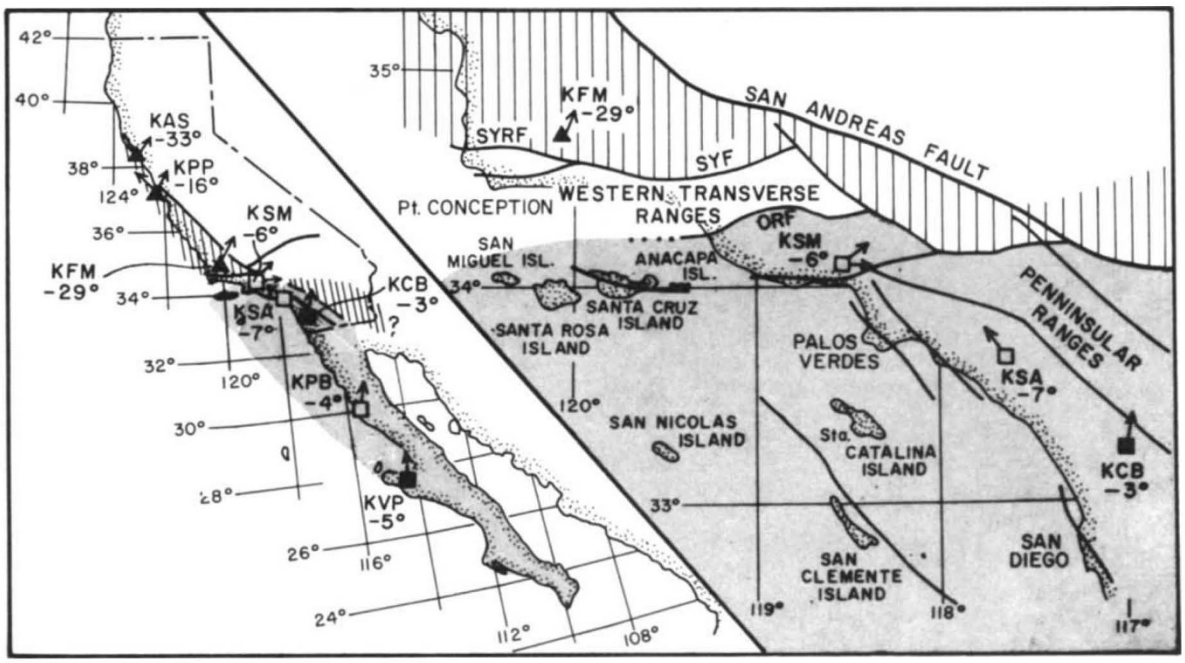


Fig. 1 Cretaceous normalized palaeolatitudes, $\Delta\theta$s, of rocks from southern and Baja California (see Table 1). □, Results from this study; ■ and ▲, previously published results (KPP[5], KFM[7], KAS[8], KCB[11], KVP[12]). □, ■, Results from the PRT; ▲, results from the SLOA. Arrows indicate directions of tectonic-corrected declinations from each locality. The similarity of $\Delta\theta$s south of the Oak Ridge Fault (ORF) and its extension into the Santa Barbara Basin suggest that this region, called the Peninsular Ranges terrane (shading), was spatially coherent in Cretaceous time. The Santa Lucia–Orocopia allochthon is shown by the vertical-lined pattern. Its southern boundary in the Western Transverse Ranges is probably the Santa Ynez (SYF) and Santa Ynez River (SYRF) faults. The normalized palaeolatitudes are also plotted against time in Fig. 3.

Table 1 Summary of palaeomagnetic results

Locality name*	KSM‖	KPB	KSA	JSA	PSA
Locality lat. (°N)	34.1	30.0	33.7	33.7	33.7
Locality long. (°E)	241.4	244.3	242.3	242.3	242.3
Age†	Tur/Maas.	Camp/Maas.	Tur./Camp.	Jurassic	Palaeocene
Confidence criteria‡	A, B, C	A, B	A, B	A, C	A, C
No. of sites§	6, 6‖	17	20	4	4
In situ inclination	37.2, 35.5	46.2	44.6	28.1	45.2
In situ declination	26.4, 104.8	−5.2	5.7	16.6	23.4
In situ α_{95}	9.9, 10.2	13.5	9.2	20.5	9.4
Corrected inclination	45.8, 48.0	44.5	45.1	11.0	43.2
Corrected declination	55.5, 90.8	3.5	−36.4	25.7	−26.0
Corrected α_{95}	7.0, 5.8	8.2	6.7	12.1	8.8
Palaeolatitude¶	28.0 ± 5.0	26.2 ± 6.5	26.6 ± 5.4	5.6 ± 6.2	25.1 ± 6.8
$\Delta\theta$¶	−6.1 ± 5.0	−3.8 ± 6.5	−7.1 ± 5.4	−28.1 ± 6.2	−8.6 ± 6.8

* KSM, Cretaceous Tuna Canyon Formation–Santa Monica Mountains; KPB, Cretaceous Punta Baja and Rosario Formations–El Rosario, Baja California; KSA, Cretaceous Ladd and Williams Formations; JSA, Jurassic Bedford Canyon Formation; and PSA, Palaeocene Silverado Formation—all from the Santa Ana Mountains. KSM, KPB and KSA sampling localities are shown in Fig. 1. JSA and PSA sampling localities are within 10 km of KSA.

† Tur., Turonian; Camp., Campanian, Maas., Maastrichtian; all late Cretaceous stages.

‡ A, Presence of reversals with antipodal polarity, B, passage of fold test, C, *in situ* direction significantly different from the present day field.

§ Four specimens were averaged per site.

‖ Two localities, each with six sites, located on opposite flanks of a large faulted anticlinal structure. Both localities have antipodal reversed sites. Agreement between two localities is markedly improved when corrected for tectonic folding (passes fold test) and declinations agree with previous estimates of large-scale rotation of the Santa Monica Mountains[15], but declinations suggest some residual differential rotation between the two localities.

¶ ±Indicates 95% confidence interval.

These results indicate that this region was about 6° farther south during late Cretaceous time relative to its present geographical position. This region, which has been termed the Peninsular Ranges terrane[9] (PRT) based on geological evidence[9,14], includes the Simi Hills, Santa Monica Mountains, Peninsular Ranges, North Channel Islands and Baja California. The SLOA north of the Santa Ynez Fault (SYF) has much larger negative $\Delta\theta$s which indicate that it was even farther south during the late Cretaceous. The region between the Oak Ridge Fault and the Santa Ynez Fault (SYF) and its extension, the Santa Ynez River Fault (SYRF), has questionable affinities.

In contrast to the negative $\Delta\theta$s of the PRT and SLOA, the $\Delta\theta$ for the North American craton was +9° during the late Cretaceous[15], indicating that the craton was about 9° farther north during this time relative to its present geographical position (Fig. 3). The 95% confidence interval ($\pm 2\sigma$) about the stable craton $\Delta\theta$ estimate for any time interval is ±8.6° (ref. 16) (Fig. 3). The PRT and SLOA $\Delta\theta$s are significantly different from the

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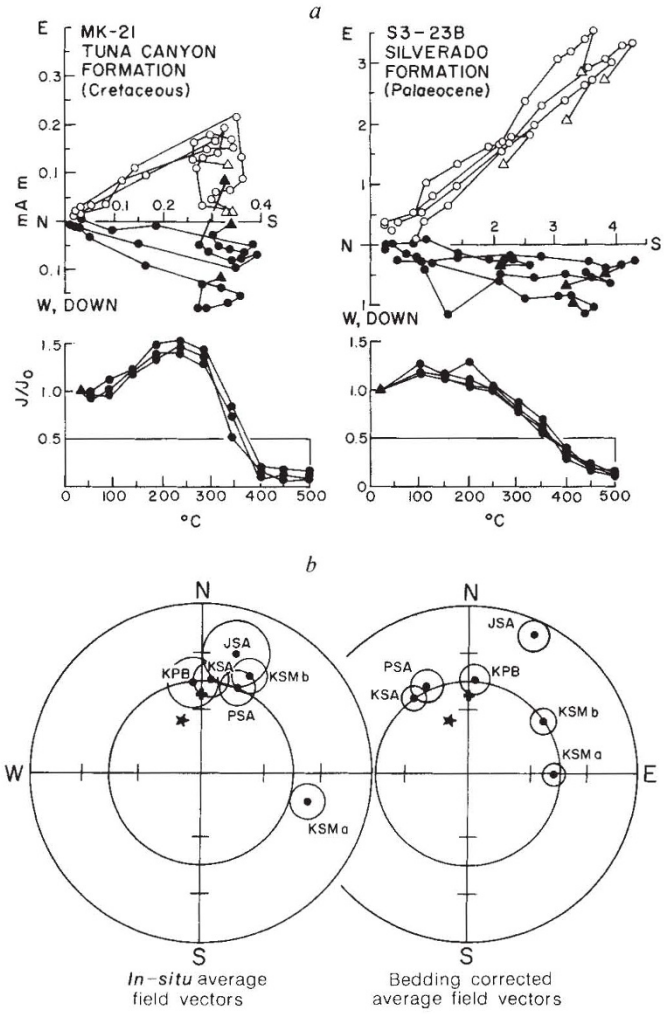


Fig. 2 Zijderveldt diagrams and stereoplots of some results from this study. *a*, Typical Zijderveldt diagrams for two sites of reversed polarity (Upper Cretaceous Tuna Canyon Formation, Palaeocene Silverado Formation) showing intensity loss and vector change with stepwise heating to 500 °C. The filled (open) circles represent the tips of NRM vectors after each heating step plotted in the horizontal (vertical) north, east (north, down) plane. *b*, *In situ* and bedding-corrected average field vectors of the localities listed in Table 1. The present axial-dipole field vector is represented by a cross. The expected late Cretaceous field vector is represented by a star. KSMa and KSMb are each the average of six sites on opposite sides of a large-anticlinal structure in the Santa Monica Mountains.

stable craton $\Delta\theta$, for they differ by $-15°$ and more than $-23°$ respectively from the stable craton result.

The distinctly different $\Delta\theta$s of the PRT, SLOA and North American stable craton indicate that they were all farther separated in latitude during the late Cretaceous than their present relative geographical positions might indicate. Less uniquely, the similarity of $\Delta\theta$s within the PRT indicate that it has remained spatially coherent since at least the late Cretaceous, with only minor re-ordering in the late Neogene due to extensional tectonics. This late Neogene tectonic regime has caused a large clockwise rotation of the Santa Monica Mountains[3,17] as noted by the rotated declinations of locality KSM (Fig. 1).

The Cretaceous $\Delta\theta$s and other Mesozoic and Cenozoic age $\Delta\theta$s (refs 5, 8, 13, 17–19) from these terranes are plotted against time in Fig. 3 to estimate each terrane's drift in palaeolatitude over the past 160 Myr. The normalized palaeolatitude, $\Delta\theta$, drift curve for the North American stable craton[15] is plotted for reference. We can see that all the $\Delta\theta$s from the PRT (squares),

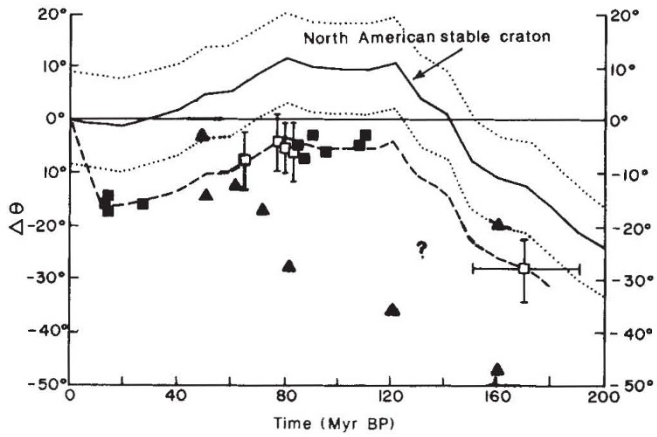


Fig. 3 Normalized palaeolatitudes of the SLOA (triangles) and PRT (squares) plotted against time. The open symbols (with 95% confidence intervals) are results from the present study, the closed symbols are previously published results[5,7,8,11–13,17–19]. The North American stable craton reference curve was derived from Irving and Irving[15] and the $\Delta\theta$ curve for the SLOA (triangles, shading) is from ref. 7. The dotted lines on either side of the stable craton reference curve indicate the 95% confidence interval ($\pm 2\sigma$) about this curve.

all older than 15 Myr, are systematically lower than the stable-craton curve by ~15°. Shifting the stable-craton curve downward 15° produces the dashed curve in Fig. 3 which not only fits all the PRT data but also connects data of different ages from within the PRT. The 95% confidence interval about the 15° offset of the PRT $\Delta\theta$ curve (dashed curve in Fig. 3) is ±6.4°, indicating that the average $\Delta\theta$s of the stable-craton and the PRT were different by more than 4σ before 15 Myr ago. The remarkable match between the offset stable-craton reference curve and these results suggests that this terrane has drifted as part of North America for most of its geological history, but that during this time it was located about 15° lower in latitude relative to the stable craton. This interpretation places the PRT along the western margin of Central America between 15 and 160 Myr ago. The PRT must have moved northward 15° since ~15 Myr ago to reach its present position relative to the stable craton.

The $\Delta\theta$s of the SLOA, on the other hand, are much lower than the PRT results, and their drift pattern[7,8] is significantly different in shape from the stable-craton reference curve. This indicates that the SLOA was not attached to the North American plate until it accreted in early Palaeogene time.

A simple history of these two terranes for the past 160 Myr can be summarized as follows: (1) location of the PRT in Central America as part of the stable craton from 160–15 Myr ago; (2) internal amalgamation of the SLOA in Cretaceous time at lower palaeolatitudes; (3) subsequent drift of the SLOA northward past and outboard from the PRT; (4) accretion of the SLOA with southern California in early Palaeogene time; (5) northward movement of the PRT along the cratonal margin since 15 Myr ago; (6) accretion of this terrane with the SLOA and North American stable craton before 5 Myr ago, (7) continued strike-slip motion along southern California faults in late Neogene time, producing the dispersed terrane pattern that we see today. The SLOA/PRT suture zone is expressed tectonically by the Transverse Ranges of southern California, and recent geophysical studies[20] support this hypothesis. The most notable elements of this history are the location of the PRT in Central America as part of the stable craton until 15 Myr ago, and its subsequent northward displacement in Neogene time. This has never been taken into account in Cenozoic tectonic reconstructions of southern and Baja California[3,21].

These results present a new view of the tectonic history of southern and Baja California and explain several perplexing aspects of the Neogene tectonic development of this region. The

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results, however, are inconsistent with some other tectonic reconstructions[21] and require further study of their possible implications.

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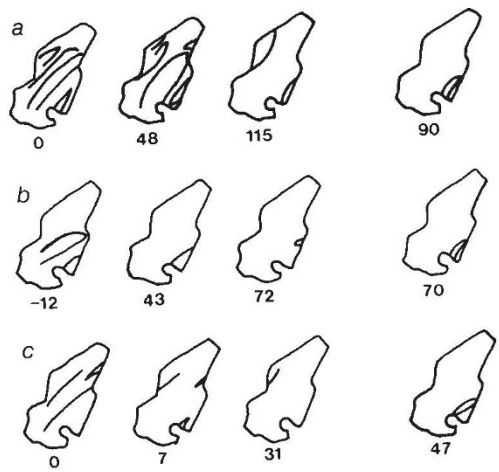


Fig. 1 Domain observations during hysteresis at room temperature (*a*), 50 °C (*b*) and 80 °C (*c*). Numbers refer to magnetic field strengths (in oersteds); positive fields point downwards. At room temperature (*a*), the particle is nearly saturated by 115 Oe and fully saturated at 193 Oe. At 50 °C (*b*), a state of near saturation occurs at 72 Oe, with full saturation at 114 Oe. At 80 °C (*c*), near saturation occurs at 31 Oe, with total saturation at 72 Oe. The final figure in each set shows the first domain pattern observed during ramp-down of the field, and the field value at which it occurred. As temperature increases there is a decrease in the external field values required for saturation and loss of saturation. The particle size is ~35 × 70 μm.

Domain observations of titanomagnetites from room temperature to Curie point and the nature of thermo-remanent magnetism in fine particles

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Single-domain theory[1] has been remarkably successful as a theoretical basis for rock magnetism and hence palaeomagnetism. However, the grain size of much of the magnetic material present in rocks is too large to be single-domain according to classical theory[2]. The resolution of this paradox may lie in the recognition that difficulties in domain wall nucleation[3-10] can preclude the multi-domain state even though it is energetically favoured. Thus, multi-domain particles are observed in a single-domain state in rocks carrying saturation remanence magnetization (IRMs). Inhibition of domain wall nucleation has also been shown to produce high-coercivity materials suitable for permanent magnets[11-13]. Thermo-remanent magnetization (TRM) is one of the most important mechanisms of remanent magnetization for the study of palaeomagnetism, being acquired for example by lavas as they cool on the Earth's surface. We present here domain studies of titanomagnetites showing the increasing importance of the magnetostatic energy in the external field as temperature is increased, and that rocks carrying weak-field TRM, like IRMs, contain multi-domain grains in a single-domain state. These particles contribute very large moments to TRM; indeed, particles of this nature may carry a substantial part of the palaeomagnetic signal. The results also suggest that weak-field cooling of dispersions of suitable fine particles might leave them in a particularly hard magnetic state of some utility.

Domain patterns in titanomagnetites of titanium content $\chi = 0.6$ were investigated during hysteresis at different temperatures and during weak-field thermal cycling, to investigate the role of domain wall nucleation in TRM. The titanomagnetites studied were from DSDP (Deep Sea Drilling Projects) basalts from Leg 49[14], and were mechanically and ionically[15] polished to remove surface stress. The Bitter pattern technique[16] was employed to observe the domain structure, using a special high-temperature colloid. The samples were placed in a brass heating stage and temperature was measured with a thermocouple. The stage was placed between the poles of an electromagnet, so that hysteresis loops could be observed, as well as thermal cycling in a variety of fields. A video camera attached to the microscope, coupled with a video recorder, allowed continuous recording of domain movement and configuration throughout each experiment.

Figure 1 shows the domain patterns resulting during hysteresis at room temperature (*a*), 50 °C (*b*) and 80 °C (*c*). Note the systematic decrease in the field required for saturation, as the temperature increases. Note also that with increasing temperature, the particle remains saturated at lower field strengths during ramp-down of the field. Figures 2 and 3 show heating and cooling cycles of the same particle in external fields of 1 Oe and 45 Oe, respectively. The particle saturates at lower temperature in the higher field. Both cycles carried the particle through its Curie point of 150 °C.

The observed domain patterns attest to the increasing dominance, at higher temperature, of the magnetostatic energy, $M_s H$, where M_s is the saturation magnetization and H the external field. This is to be expected, as other terms are dependent on M_s to higher powers (for example, the demagnetization energy is proportional to M_s^2). As M_s decreases with increasing temperature, the linear term becomes dominant.

No substantial increase in the number of domain walls is seen with increasing temperature. This possibility had been considered because the temperature dependence of the various energy terms should allow nucleation to occur more easily. Neither in these samples, nor in larger particles with numerous walls, have we seen a substantial increase in the number of walls as the sample was heated. This is again consistent with the increasing importance of the magnetostatic energy in the external field relative to the demagnetizing energy.

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